

The Raspberry Pi is Fuelling a D-I-Y Revolution

Priced easy on the pocket and being small enough to fit into it, the Raspberry Pi is truly creating a revolution in hands-on learning and lunchroom innovation. Students, hobbyists and even scientists are grabbing the Pi and doing some amazing things with it – devices that can feed cats, water plants, guard the home, monitor pollution, improve food production and more. People are carrying it around in funky cases, using it as a computer or to add more snazzy features to devices they see around them. Here are some such Raspberry Pi-based open source innovations that you could hack into too...

Complete ecosystem for the Internet of Things

<http://pinocc.io/>



Pinoccio is a complete, open source hardware platform for the Internet of Things (IoT). Developed by Eric Jennings and Sally Carson, and launched in March 2013, Pinoccio is a wireless, Web-ready microcontroller with Wi-Fi, a lithium-polymer battery, a built-in radio and the required application programming interface (API). The board is Arduino-compatible, so you can use the Arduino IDE too for programming it. Pinoccio will help you to easily develop smart devices for the IoT era, where everything from public infrastructure to household gadgets is becoming connected and intelligent.

The open twist: Pinoccio is completely open source, and the designs and code are available on Github (<https://github.com/Pinoccio>). Like the Arduino Mega, this board is also based on the ATmega128RFA1, but it has a much smaller footprint and also includes a built-in radio. All firmware files including bootloaders and app hex files; hardware reference files including schematics, board layout and datasheets; Wi-Fi hardware references and shield references are available online. The Eagle library is also available with landing patterns and silkscreens of devices

used on the Pinoccio boards. The board features 256 K Flash, 32K SRAM and 8K EEPROM; hardware support for wireless, over-the-air programming, 2.4GHz connectivity using the 802.15.4 wireless standard and Web connectivity via an available Wi-Fi shield. It also has sufficient I/O pins and ports including a micro USB port for charging and programming, a Li-Po rechargeable battery (550mAh) and the ability to check battery voltage from the microcontroller via a low-power MOSFET-based circuit.

Water the plants just right, with H2O IQ

<http://blog.valkyriesavage.com/blog/2013/01/18/h2o-iq/>

Here is an interesting device born out of a design course project! When a group of students taking a course at the Citrus Invention Lab, University of California, Berkeley, had to invent a smart device as part of their course, they decided to build H2O IQ, which could automatically water plants in a



garden. The goal was to build a device that could understand the moisture requirements of a species and provide just enough water at just the required times, so that the plant would be healthy and, at the same time, water would be conserved. However, when they discussed their project with